BCI Inefficiency: Does Pre-Trial Mu Power Predict Motor-Imagery Decoding Performance?

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***Abstract*—A sizeable fraction of users cannot operate a motor-imagery (MI) brain–computer interface (BCI), and today this is discovered only after a long calibration session. We tested whether the power of the sensorimotor mu rhythm (8–13 Hz) in the second before each trial predicts how well a subject's MI can be decoded. Using all 52 subjects of the public Cho2017 left/right-hand MI dataset, we band-pass filtered (1–60 Hz) and re-referenced (common average) the EEG, verified contralateral mu/beta event-related desynchronization (ERD), and decoded left versus right imagery from 52 log band-power features with logistic regression and shrinkage LDA. Evaluation was strictly cross-subject (`GroupKFold` by subject, nested tuning). Both decoders reached ROC AUC 0.65 ± 0.03 (accuracy 0.61), well above a permutation chance level of 0.50 ($p = 0.0033$). Per-subject AUC ranged from 0.43 to 0.985, and per-subject pre-trial log mu power correlated with it (Spearman $\rho = 0.43$, $p = 0.0016$, $N = 52$), supporting the hypothesis. Follow-up analyses showed that the correlation persists with per-subject feature normalization ($\rho = 0.38$) and with within-subject decoders ($\rho = 0.41$), and that a predictor computed from a separate one-minute resting recording, the height of the mu peak above the $1/f$ background, correlates even more strongly ($\rho = 0.55$, $p = 2 \times 10^{-5}$), close to the $r = 0.53$ reported by Blankertz et al.**

I. Introduction

Motor-imagery BCIs translate the imagined movement of a body part into a command by detecting changes of sensorimotor rhythms in the EEG. Their performance varies enormously between people: an estimated 15–30% of users never reach reliable control, a phenomenon called BCI illiteracy or BCI inefficiency [1]. Because a user's suitability is only revealed after a calibration session, a cheap neurophysiological predictor measured *before* training would let a system decide early who needs extra training, a different paradigm, or no MI-BCI at all.

Blankertz et al. [2] showed that the strength of the sensorimotor rhythm in a two-minute eyes-open resting recording correlated with later BCI feedback performance at $r = 0.53$ in 80 users. Wang et al. [3] confirmed on 105 subjects that resting-state alpha-band features are among the most informative predictors of MI performance, and that the result depends on more than absolute power. Geronimo et al. [4] used mu amplitude at the start of individual trials to predict trial-level success, which is closest to our pre-trial window, but they studied variation within subjects rather than differences between them.

Hypothesis. *Subjects with higher pre-trial mu power achieve better MI decoding performance.* The physiological rationale (Section II) is that a strong idle mu rhythm leaves more power to suppress during imagery, producing a larger ERD and a clearer left–right contrast.

We test this on the 52 subjects of Cho2017 [5] with a leakage-free cross-subject pipeline built only from the methods of the course notebooks 00–06 (MNE-Python [7], MOABB [8], scikit-learn).

II. Background

Rhythms. Alpha (8–13 Hz) is strongest over occipital cortex with the eyes closed. The *mu* rhythm shares this band but is generated over the sensorimotor cortex (C3, Cz, C4); it reflects an idle motor system and is suppressed by executing, imagining or observing a movement. *Beta* (13–30 Hz) also decreases during movement and rebounds above baseline after it ends (post-movement beta rebound). Gamma (> 30 Hz) accompanies active processing, but on the scalp it is dominated by muscle (EMG) activity, so our features stay in 8–30 Hz.

ERD/ERS. Event-related desynchronization and synchronization quantify band power during a task relative to a baseline [6]:

$$\mathrm{ERD/ERS}\,(\%) = \frac{P_{\text{task}} - P_{\text{baseline}}}{P_{\text{baseline}}} \times 100, \tag{1}$$

where P_{task} is the band power (area under the PSD over the band) during imagery and P_{baseline} the band power in the same channel before the cue. Negative values (ERD) indicate that a population of neurons stops oscillating in synchrony as the area becomes engaged; positive values (ERS) indicate increased synchrony, such as the beta rebound. Fig. 1 illustrates both quantities.

Contralateral organization. Each hemisphere controls the opposite hand, so right-hand imagery should produce the stronger ERD over C3 (left hemisphere) and left-hand imagery over C4 (right hemisphere). This asymmetry is what a left/right decoder exploits.

III. Dataset

Cho2017 [5] was loaded with MOABB. Table I summarizes the audit. Each trial shows a fixation cross for 2 s, then the cue "left hand" or "right hand" for 3 s of imagery, then a blank screen and a 4.1–4.8 s break (Fig. 3). MOABB demeans each trial and concatenates trials with zero padding, so we only analyse epochs inside trials. Fig. 2 shows the sensor layout and a sample of the recording. The original `.mat` files additionally contain a ≈66 s resting recording that MOABB does not load; we use it only in the follow-up analysis (Section VI).

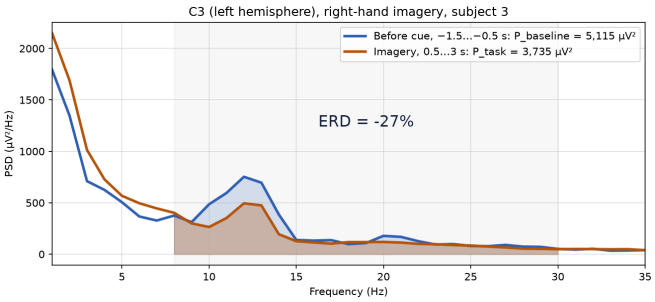


Fig. 1. Median Welch PSD at C3 over 100 right-hand trials of subject 3, before the cue (P_{baseline}, blue) and during imagery (P_{task}, orange). Shaded areas are the 8–30 Hz band powers; ERD computed from these median spectra is -27% (the median of per-trial ERD reported in Section V is -46%).

TABLE I
DATA AUDIT OF CHO2017

Subjects	52 (IDs 1–52), all used
Sessions / runs	1 session; 5 runs (6 for s7, s9, s46)
Channels	64 EEG (10-10, Biosemi) + 4 EMG + stim
Sampling rate	512 Hz
Units	V
Reference	not reported; we apply CAR
Events	1 = left hand, 2 = right hand (cue)
Trials	100 per class (120 for s7, s9, s46)
Total	10 520 trials
Epochs	-1.5 to 2.998 s, 2304 samples

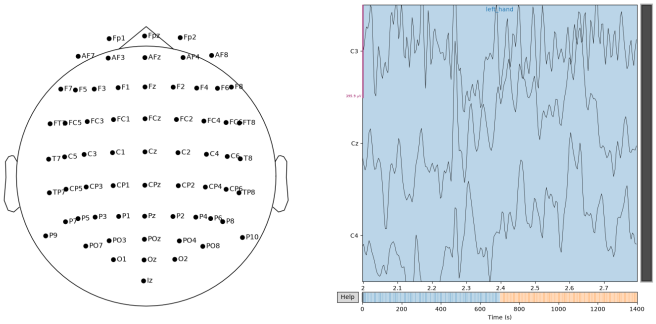


Fig. 2. Left: layout of the 64 EEG sensors (10-10 system). Right: C3, Cz and C4 of subject 3 after filtering and CAR, 0.8 s from the first cue.

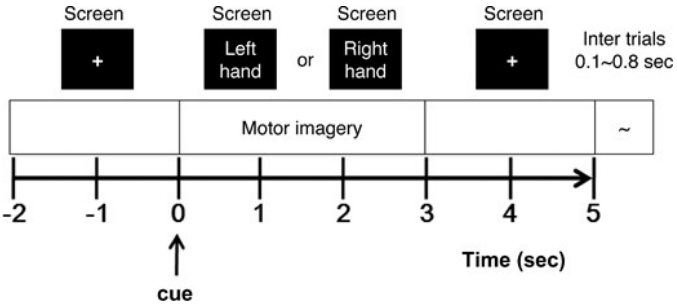


Fig. 3. Trial timeline of Cho2017 (from [5]). Our pre-trial window is -1.5 to -0.5 s (fixation), the feature window 0.5 to 2.5 s (imagery).

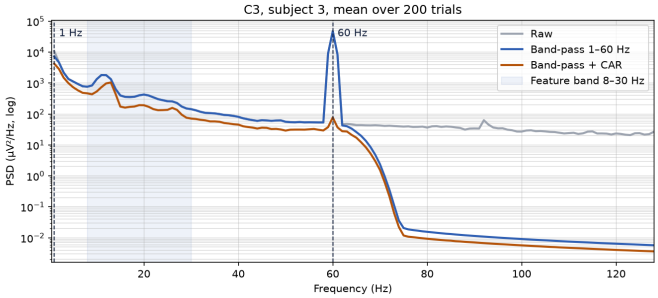


Fig. 4. PSD of C3 (subject 3, mean over 200 trials, log scale) for the raw signal, after the 1–60 Hz band-pass and after CAR. The raw curve is hidden below 60 Hz, where the band-pass leaves it unchanged; note the 60 Hz line peak removed by CAR.

IV. METHODS

A. Preprocessing

For every subject the continuous EEG was filtered with a 1–60 Hz zero-phase FIR band-pass (Hamming window, MNE defaults: 1 Hz lower and 15 Hz upper transition band). The high-pass removes the large DC offset of the raw signal (mean of C3 ≈ -35 mV in subject 3) while keeping mu and beta. We then applied a common average reference (CAR) over the 64 EEG channels. In subject 3 CAR reduced the C3–C4 correlation of the first trial from 0.845 to -0.140 and the mean correlation of seven central channels from 0.888 to 0.374, removing a large oscillation shared by all channels (Fig. 5). CAR does not decorrelate channels: neighbours such as C3 and FC3 remain correlated because they record the same cortical area (Fig. 6). Epochs spanned -1.5 to 2.998 s around the cue, without baseline correction.

Fig. 4 shows the PSD of C3 before filtering, after the band-pass and after CAR (computed for this report with the same steps). The band-pass removes power above ≈70 Hz by more than three orders of magnitude and leaves the 8–30 Hz feature band unchanged. It also reveals a strong 60 Hz power-line peak (the data were recorded in Korea, where the mains frequency is 60 Hz). Because our upper passband edge is exactly 60 Hz, the filter does not attenuate it; CAR does, by more than two orders of magnitude, since line noise is nearly identical on all channels. The peak lies far from our features, but a 60 Hz notch or a lower cut-off (e.g. 40 Hz) would be the cleaner choice and would also make amplitude-based artifact screening less sensitive to line noise. No resampling was applied: 512 Hz is far above twice the highest frequency of interest.

B. Artifact screening

We computed the peak-to-peak (PTP) amplitude of each epoch (0–3 s) and flagged epochs whose maximum PTP over channels exceeded the median $+3\times$MAD. In subject 3, 23 of 200 epochs were flagged (12 left, 11 right); in 20 of them the worst channel was Fp1 or Fp2 and in 3 AF7 or AF8, consistent with eye blinks, whereas PTP at C3/C4 stayed narrow (Fig. 7). However, the five epochs with the largest PTP at C3 (82, 176, 2, 166, 117) and at C4 (117, 92, 141, 2, 37) all appear in the frontal outlier list: a blink of several millivolts at Fp1/Fp2

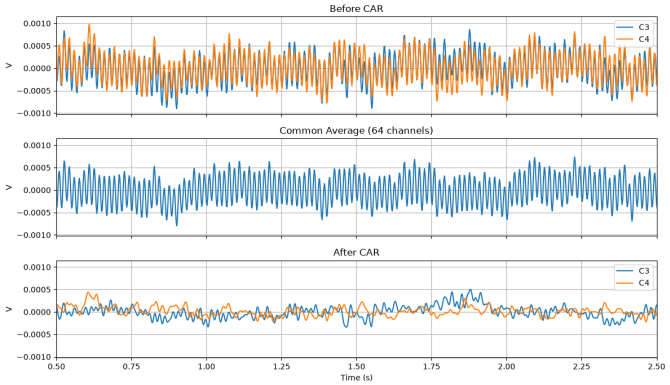


Fig. 5. C3 and C4 before CAR (top), the 64-channel common average (middle) and after CAR (bottom); subject 3, first left-hand trial.

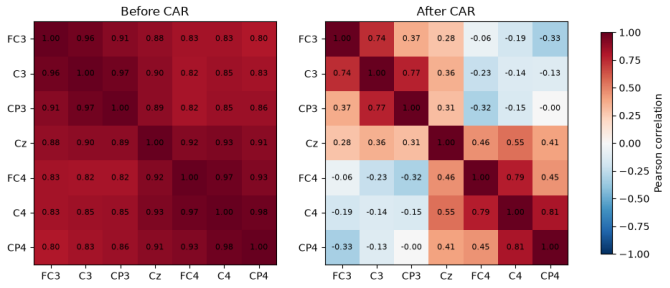


Fig. 6. Pearson correlations between seven central channels before and after CAR (subject 3, first left-hand trial).

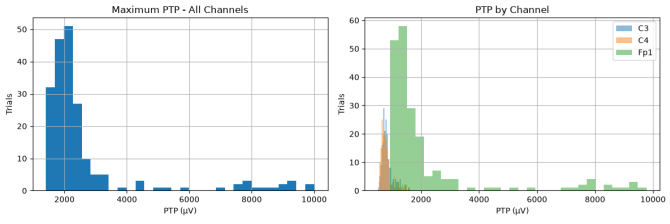


Fig. 7. Peak-to-peak amplitude per epoch (subject 3): maximum over all channels (left) and per channel for C3, C4 and Fp1 (right).

shifts the common average, and CAR subtracts that shift from every channel, so the artifact reaches C3 and C4 (Fig. 8). We deliberately rejected *no* epochs in the main analysis: removing difficult trials from test folds would inflate performance, and the features use only central channels. The effect of rejection on training data is tested in Section VI.

C. Features

For each trial, Welch PSD (1 s Hann windows, 50% overlap) was computed in the 0.5–2.5 s window on 13 sensorimotor channels (FC3, FCz, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CPz, CP4) and integrated (trapezoid rule) over four bands: 8–10, 10–13, 13–20 and 20–30 Hz. The logarithm of band power in μV^2 gives $13 \times 4 = 52$ features. These per-trial features have no fitted parameters, so computing them before cross-validation causes no leakage.

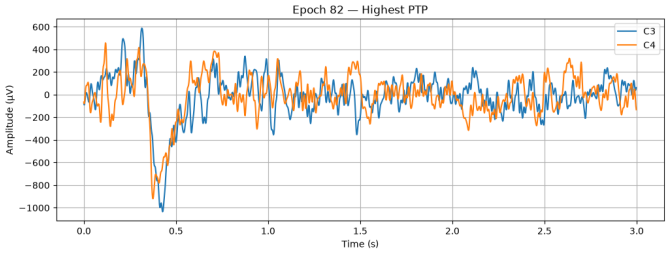


Fig. 8. Epoch 82 of subject 3, the largest PTP overall: a deflection near 0.3–0.5 s appears on both C3 and C4 after CAR; its worst channel is Fp2.

D. Classifiers and evaluation

Two scikit-learn pipelines were compared: `StandardScaler` + logistic regression and `StandardScaler` + LDA with Ledoit–Wolf shrinkage (`solver="lsqr"`, `shrinkage="auto"`). Evaluation used `GroupKFold(4)` grouped by subject, so about 39 subjects train and 13 unseen subjects test in every fold. The regularization $C \in \{0.001, 0.01, 0.1, 1, 10\}$ of logistic regression was tuned by an inner `GroupKFold(3)` on the training subjects only. We report ROC AUC and accuracy as mean $\pm$ SD over folds (classes are balanced, so accuracy equals balanced accuracy) and per-subject AUC from out-of-fold predictions. Chance level was estimated with 300 label permutations within subjects (`permutation_test_score`, same grouped CV, seed 42).

E. Hypothesis test

For each subject, pre-trial mu power was computed as Welch power in 8–13 Hz in the -1.5 to -0.5 s window (fixation, before the cue), log-transformed and averaged over the 13 channels and all trials. The pre-trial and feature windows do not overlap. We correlated this value with the subject's held-out AUC using Spearman's rank correlation ($N = 52$). The two classifiers were compared with a paired Wilcoxon signed-rank test across subjects.

F. Choice of statistics

We used rank-based tests throughout. Per-subject AUCs are bounded, skewed and contain a few very high values (up to 0.985), so Spearman's ρ is more robust than Pearson's r to single extreme subjects and only assumes a monotonic relation. The permutation test shuffles labels within each subject and repeats the entire grouped cross-validation, so the null distribution keeps the subject structure and the class balance of the real data; with 300 permutations the smallest attainable p is $1/301$. Classifier and pipeline comparisons are paired by subject, which the Wilcoxon signed-rank test respects without assuming normally distributed differences. All tests are two-sided at $\alpha = 0.05$; the unit of analysis for every between-subject statement is the subject ($N = 52$), never the trial.

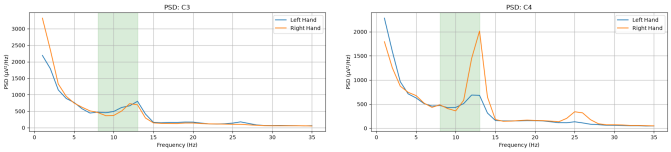


Fig. 9. Mean PSD at C3 (left) and C4 (right) for left- and right-hand imagery, subject 3; mu band shaded.

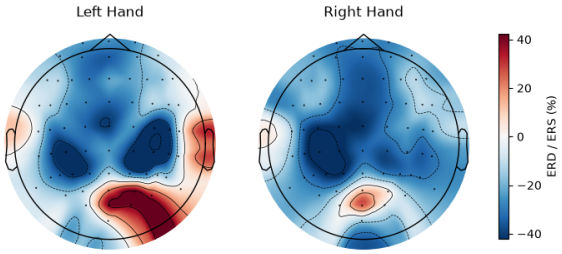


Fig. 10. ERD/ERS (%) topographies, 8–30 Hz, task 0.5–2.5 s vs. pre-trial, median over trials, subject 3. Blue: power decrease.

TABLE II
CROSS-SUBJECT DECODING (GROUPKFOLD(4) BY SUBJECT)

Model	ROC AUC	Accuracy
Logistic regression	0.654 ± 0.033	0.609 ± 0.025
Shrinkage LDA	0.652 ± 0.033	0.608 ± 0.027
Permutation chance	0.499	–

Fold AUCs (LogReg): 0.641, 0.618, 0.663, 0.695.

V. RESULTS

A. Neurophysiological check

Averaged over 100 trials per class, subject 3 showed lower mu power at C3 during right-hand imagery (2552 vs. 2864 μV^2) and at C4 during left-hand imagery (2632 vs. 4039 μV^2) (Fig. 9). Relative to the pre-trial window, median 8–30 Hz ERD was -56% at C4 and -28% at C3 for the left hand, and -46% at C3 and -22% at C4 for the right hand: both hemispheres desynchronize, the contralateral one about twice as much. Topographic maps over all 64 channels show the same contralateral pattern (Fig. 10).

B. Decoding

Table II lists cross-subject performance. The nested search chose $C = 0.01$ in all four folds. The permutation distribution was centred at AUC 0.499; no permutation reached the observed 0.652, giving $p = 1/301 = 0.0033$, the smallest value attainable with 300 permutations (Fig. 12). Logistic regression outperformed LDA by 0.003 AUC per subject on average (Wilcoxon $p = 0.043$), a difference far below the fold-to-fold SD and of no practical relevance. ROC curves of the four test folds are shown in Fig. 11; fold-to-fold variation reflects which 13 subjects form the test set.

C. Hypothesis test

Per-subject held-out AUC ranged from 0.43 to 0.985; five subjects were below 0.5 and 20 of 52 (LDA: 19) below 0.6.

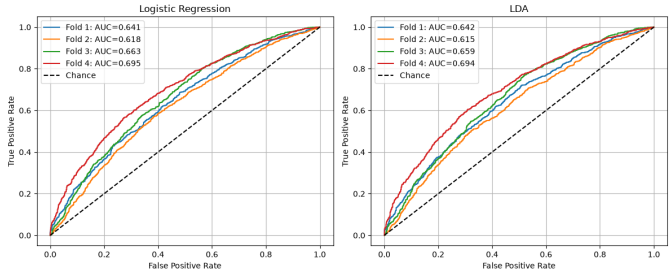


Fig. 11. ROC curves of the four subject-wise test folds for logistic regression (left) and LDA (right).

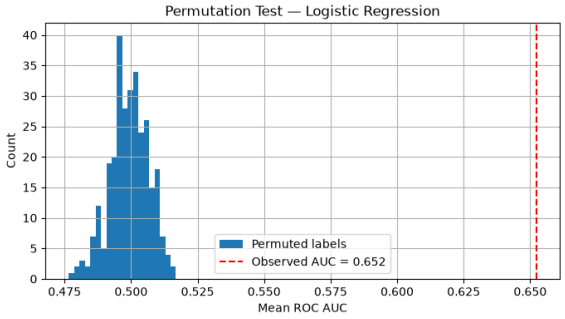


Fig. 12. Permutation test: AUC with labels shuffled within subjects (300 permutations) and the observed AUC.

Pre-trial log mu power correlated positively with AUC: $\rho = 0.428$, $p = 0.0016$ for logistic regression and $\rho = 0.435$, $p = 0.0013$ for LDA (Fig. 13). The hypothesis is supported in this setting. Table III lists every subject.

VI. FOLLOW-UP ANALYSES

Our main design leaves an alternative explanation open: the decoder is trained across subjects on *absolute* log power, so subjects with high mu power might simply match the population model better, independently of how separable their own imagery is. We therefore ran four follow-up analyses in a separate notebook (`recommendations/`), each changing one factor relative to the main pipeline. All use the same 52 features and shrinkage LDA; the first step reproduced the main result exactly ($\rho = 0.428$ LogReg, 0.435 LDA).

Methods. (i) *Within-subject decoding*: one LDA per subject, evaluated leave-one-run-out (5 runs; 6 for s7, s9, s46), so runs never mix between training and test. (ii) *Per-subject normalization*: each subject's features were z-scored with that subject's own mean and SD (label-free) before the same cross-subject `GroupKFold(4)`. (iii) *Alternative predictors*: relative pre-trial mu (8–13 Hz divided by 4–35 Hz power), the median task ERD of mu over the 13 channels, and three measures from the ≈66 s resting recording in the original Cho2017 files, after the same filter and CAR: log mu power, relative mu power, and the height (dB) of the 8–13 Hz peak above a $1/f$ fit at C3 and C4, a simplified version of the predictor of [2]. Spearman correlations are reported with 95% bootstrap confidence intervals (2000 resamples over subjects).

TABLE III
PER-SUBJECT PRE-TRIAL LOG MU POWER (LOG μV^2) AND HELD-OUT ROC AUC: CROSS-SUBJECT LOGISTIC REGRESSION (LR) AND LDA FROM THE MAIN ANALYSIS, WITHIN-SUBJECT LDA (W) FROM THE FOLLOW-UP

ID	Mu	LR	LDA	W	ID	Mu	LR	LDA	W	ID	Mu	LR	LDA	W	ID	Mu	LR	LDA	W
1	9.48	0.700	0.702	0.773	14	9.70	0.985	0.984	0.995	27	7.26	0.499	0.491	0.532	40	8.28	0.446	0.461	0.562
2	7.90	0.553	0.560	0.512	15	7.81	0.754	0.755	0.803	28	7.78	0.454	0.453	0.422	41	9.20	0.929	0.927	0.910
3	8.31	0.952	0.947	0.979	16	8.42	0.699	0.685	0.652	29	6.89	0.566	0.569	0.463	42	7.39	0.714	0.699	0.703
4	9.09	0.784	0.775	0.842	17	7.07	0.430	0.426	0.484	30	7.18	0.701	0.694	0.683	43	8.19	0.977	0.971	0.985
5	9.41	0.652	0.639	0.700	18	7.48	0.696	0.699	0.767	31	8.44	0.691	0.652	0.817	44	8.90	0.675	0.651	0.748
6	7.73	0.584	0.583	0.679	19	7.99	0.670	0.664	0.645	32	7.77	0.539	0.539	0.604	45	8.96	0.655	0.662	0.618
7	7.64	0.610	0.607	0.612	20	8.89	0.666	0.667	0.616	33	8.68	0.616	0.615	0.571	46	8.84	0.663	0.673	0.766
8	7.79	0.543	0.536	0.521	21	8.66	0.576	0.582	0.784	34	7.59	0.522	0.524	0.468	47	9.12	0.722	0.718	0.707
9	8.31	0.754	0.744	0.673	22	8.34	0.740	0.752	0.744	35	7.38	0.483	0.493	0.724	48	9.66	0.887	0.882	0.878
10	8.63	0.739	0.749	0.774	23	8.69	0.762	0.745	0.849	36	8.37	0.694	0.676	0.740	49	8.45	0.599	0.604	0.761
11	9.10	0.557	0.562	0.619	24	7.61	0.787	0.782	0.769	37	7.58	0.513	0.511	0.756	50	8.40	0.565	0.562	0.558
12	9.59	0.706	0.708	0.671	25	7.99	0.745	0.751	0.716	38	7.62	0.539	0.527	0.492	51	8.18	0.595	0.594	0.655
13	8.01	0.557	0.542	0.584	26	8.52	0.558	0.555	0.674	39	9.50	0.637	0.637	0.638	52	8.13	0.736	0.732	0.670

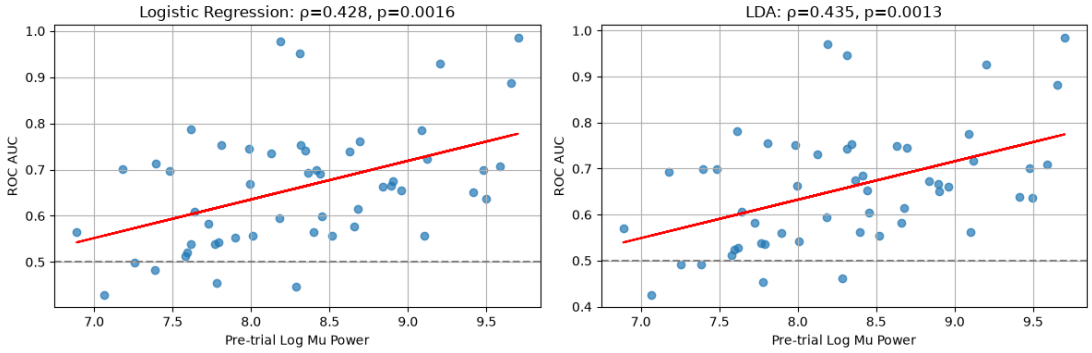


Fig. 13. Per-subject pre-trial log mu power vs. held-out ROC AUC of the cross-subject decoders ($N = 52$). Red: least-squares trend; dashed: chance.

(iv) *Rejection ablation*: trials whose maximum PTP exceeded the subject's median $+3\times$MAD were removed from *training* folds only; all test trials were still scored.

Results. Table IV and Figs. 14–16 summarize the outcome. The correlation between pre-trial mu and AUC remained with normalized features ($\rho = 0.38$, $p = 0.005$) and with within-subject decoders ($\rho = 0.41$, $p = 0.003$). Within-subject decoders were better than the cross-subject one (0.690 vs. 0.658, Wilcoxon $p = 0.016$; 12 instead of 19 subjects below 0.6), but the ranking of subjects was largely preserved ($\rho = 0.75$ between the two). Normalization (+0.006, $p = 0.19$) and training-fold rejection of 860 trials (8.2%; +0.0002, $p = 0.59$) did not change performance.

Pre-trial and resting log mu were strongly correlated across subjects ($\rho = 0.82$), and resting-state predictors were at least as informative as the pre-trial one: rest log mu $\rho = 0.48$, rest relative mu $\rho = 0.45$ and rest SMR peak $\rho = 0.55$ ($p = 2\times10^{-5}$) for the cross-subject decoder (Fig. 15). Relative pre-trial mu was weaker but still positive ($\rho = 0.34$, $p = 0.013$). The crude task ERD measure did not correlate with AUC ($\rho = -0.19$, $p = 0.18$).

VII. DISCUSSION

The hypothesis is supported. Across 52 subjects, those with stronger mu power before the cue were decoded better,

TABLE IV
FOLLOW-UP DECODERS (SHRINKAGE LDA, 52 SUBJECTS)

Decoder	AUC	< 0.6	ρ with mu
Cross-subject (main)	0.658	19	0.44 ($p = 0.001$)
+ z-scored per subject	0.664	18	0.38 ($p = 0.005$)
+ training rejection	0.658	19	0.43 ($p = 0.002$)
Within-subject	0.690	12	0.41 ($p = 0.003$)

AUC: mean of per-subject held-out AUCs; < 0.6: number of subjects.

with a moderate rank correlation of about 0.4. The effect does not hinge on the cross-subject design: it survives per-subject normalization and appears with each subject's own decoder. It is also not only a matter of overall signal amplitude (skull thickness, electrode gain), because relative mu, which cancels broadband power, still correlates.

Relation to prior work. The pre-trial measure behaves like a stable trait: it correlates at 0.82 with the same quantity in a separate resting recording, and the resting SMR-peak predictor reproduces the effect size of Blankertz et al. [2] ($\rho = 0.55$ vs. $r = 0.53$) on an independent dataset. This agrees with Wang et al. [3], who found resting alpha-band features most informative and noted that more than absolute power matters. Unlike Geronimo et al. [4], who used trial-start mu to predict single-trial success within subjects, our result

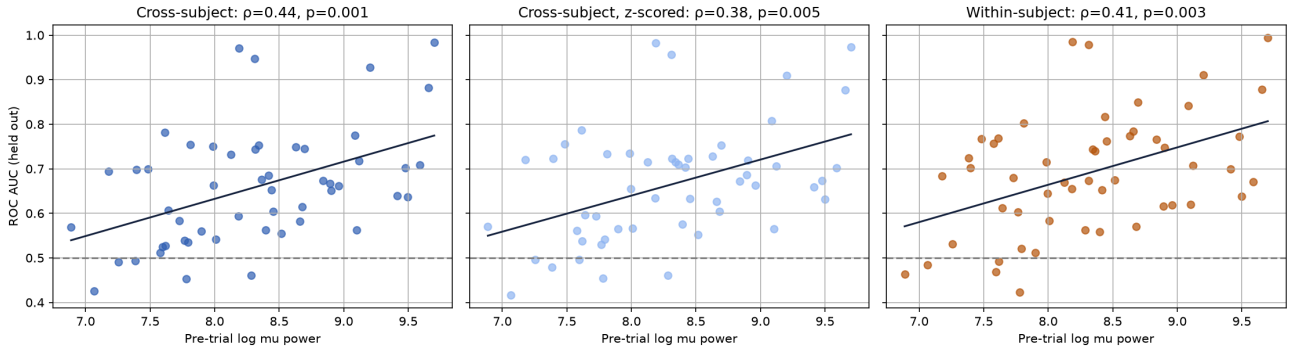


Fig. 14. Pre-trial log mu power vs. held-out AUC for the cross-subject, per-subject z-scored and within-subject decoders.

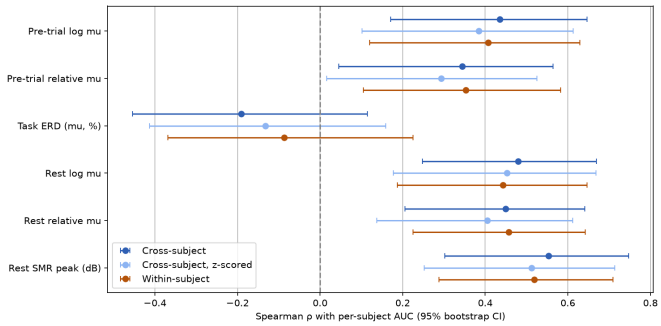


Fig. 15. Spearman ρ (95% bootstrap CI) between six per-subject predictors and AUC for three decoders.

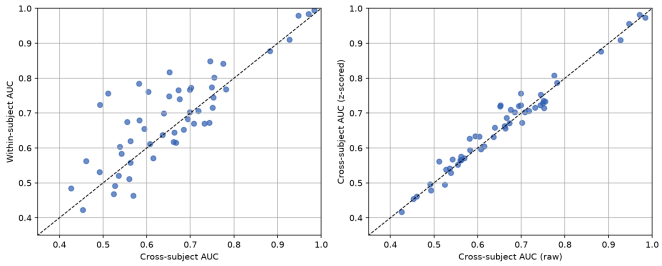


Fig. 16. Per-subject AUC: within- vs. cross-subject (left) and z-scored vs. raw cross-subject features (right).

concerns differences between subjects.

Limitations. (1) The analysis is correlational and explains about a fifth of the rank variance ($\rho^2 \approx 0.19$ for the main predictor); several high-mu subjects still decode poorly and vice versa. (2) The pre-trial window lies in the fixation period of the same session, so anticipation of the cue could contribute. (3) Many correlations were computed in the follow-up (18 predictor–decoder pairs); with a Bonferroni threshold of $0.05/18 = 0.0028$, the pre-trial result for the cross-subject and within-subject decoders ($p = 0.0013, 0.0027$) and all three rest SMR-peak results remain significant, whereas the z-scored pre-trial result ($p = 0.005$) does not. The follow-up should be read as exploratory. (4) Our ERD measure, a median over 13 channels and both classes, ignores the contralateral pattern and is too crude to judge whether ERD magnitude itself predicts performance. (5) One dataset, offline analysis, and one session per subject; the decoder never adapted to the user, so the results say nothing about who can learn with feedback. (6) Cross-subject AUC of 0.65 is modest and well below what a calibrated online BCI needs. (7) All single-subject figures use subject 3, which turned out to be atypical: it is the third-best decoder of 52 (AUC 0.952, median 0.665), although its pre-trial mu power (8.31) is exactly the group median. The clean contralateral ERD in Figs. 9–10 is therefore a best case, and the subject itself illustrates that mu power explains only part of who decodes well. (8) Two preprocessing side effects remain: the 60 Hz line peak passes the band-pass and is removed only by CAR, and CAR in turn spreads large frontal blink artifacts to C3 and C4 (Fig. 8); neither touches the 8–30 Hz features much, but both inflate amplitude-based artifact screening.

Recommendations. Based on these results we would change four things in a follow-up study: (i) add a 60 Hz notch (or a 40 Hz low-pass) and screen artifacts on a reference that excludes the frontal channels; (ii) use the resting recording as the primary predictor, since it is cheap, independent of the task and the strongest predictor we found; (iii) test ERD magnitude with a class-specific contralateral measure (C3 for the right hand, C4 for the left) instead of a pooled median; and (iv) pre-register the predictor and validate it prospectively on new users with online feedback, which is the setting in which a screening tool would actually be used.

Reproducibility. Package versions: Python 3.11.2, MNE 1.13.2, MOABB 1.7.2, scikit-learn 1.9.1, SciPy 1.17.1, NumPy 2.4.6. `GroupKFold` is deterministic; permutations and bootstraps use seed 42. Because the full dataset is ≈10 GB, a mini notebook runs the identical pipeline on 12 randomly drawn subjects (seed 42, subject 3 always included) within the Colab budget; processing took 107 s after download, and setting `N_SUBJECTS = 52` reproduces the full analysis. On this subset the decoder reached AUC 0.62 ± 0.10 (permutation $p = 0.0033$) and the correlation had the same sign and size ($\rho = 0.52$ LogReg, 0.47 LDA) but was not significant ($p = 0.08, 0.12$), as expected with $N = 12$: the full 52-subject analysis is needed for the hypothesis test.

VIII. Conclusion

Contralateral mu/beta ERD was clearly present in Cho2017, a leakage-free cross-subject decoder of left- vs. right-hand imagery reached AUC 0.65 (permutation $p = 0.0033$), and subjects with higher pre-trial mu power were decoded better ($\rho = 0.43$). The relation persisted with normalized features and with per-subject decoders, and a one-minute resting recording predicted performance at least as well ($\rho = 0.55$). A short mu-rhythm measurement before training could therefore help flag likely low performers before a long calibration session, although it explains only part of the variance and should be validated prospectively with online feedback.

Appendix: AI tools and contributions

Claude Code (Anthropic) was used to draft the presentation slides and this report from the notebook outputs, to review the pipeline, and to implement the follow-up analyses in a separate folder (`recommendations/`); all numbers come from executed notebooks. ChatGPT (OpenAI) was used mainly to study the software libraries (MNE-Python, MOABB, scikit-learn) and the methods taught in the course notebooks. No AI tool was used to invent results, citations or dataset facts.

Roles (each member leads one area and reviews another; with three members the data and decoding roles are merged): **Alar Akilbekov**, data and decoding lead (loading, audit, event coding, features, classifier pipeline, cross-validation), reviews evaluation; **Aruzhan Kaltay**, signal processing lead (filtering, re-referencing, artifact screening, PSD/ERD check), reviews the data audit; **Alim Daniyarov**, evaluation and reporting lead (chance level, statistical tests, follow-up analyses, report and slides), reviews decoding and signal processing.

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